



EXPLORING RECOMMENDATIONS SYSTEMS

MOVIE

▶ Play

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Who's Watching?



Leonard



Meron



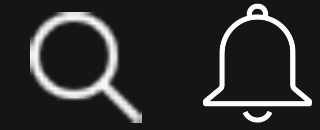
Diego



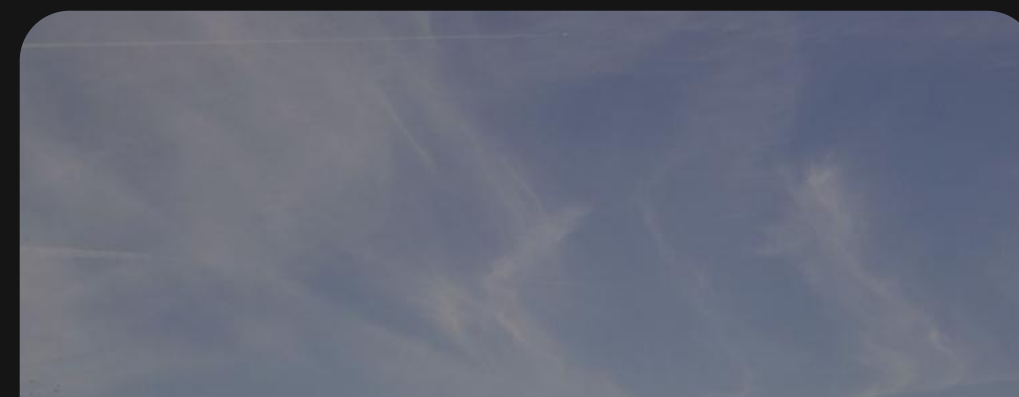
Mobo

Start

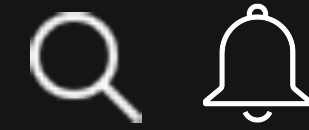
MOVIE



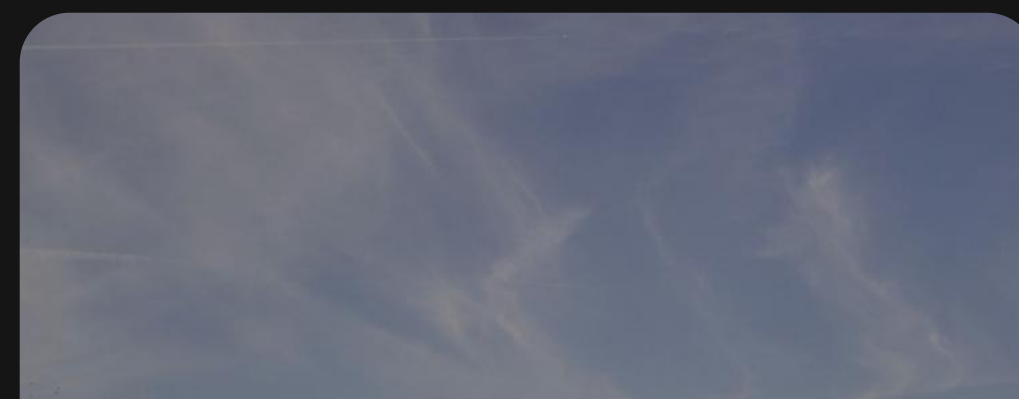
How do they
work...



MOVIE



Recommendations
seems repetitive...





This is why we decided to explore how Recommendation system works ... in practice 🎬



No. 1

The world of
recommendation
systems - In a Nutshell



No. 2

SLi-Rec Model as our go
to choice



No. 3

Our experiments and
main findings

RECOMENDATIONS

How do they work?

1. Commonly Picked

2. Time Sensitive

3. Other factors

#1 Question

James watches The Simpson every morning. On weekends, on an irregular schedule, he would watch fantasy movies. What would be recommended to him at 10 am on a Thursday?

a. The Simpson

b. The Lord of the Rings

c. Jumanji

d. Harry Potter

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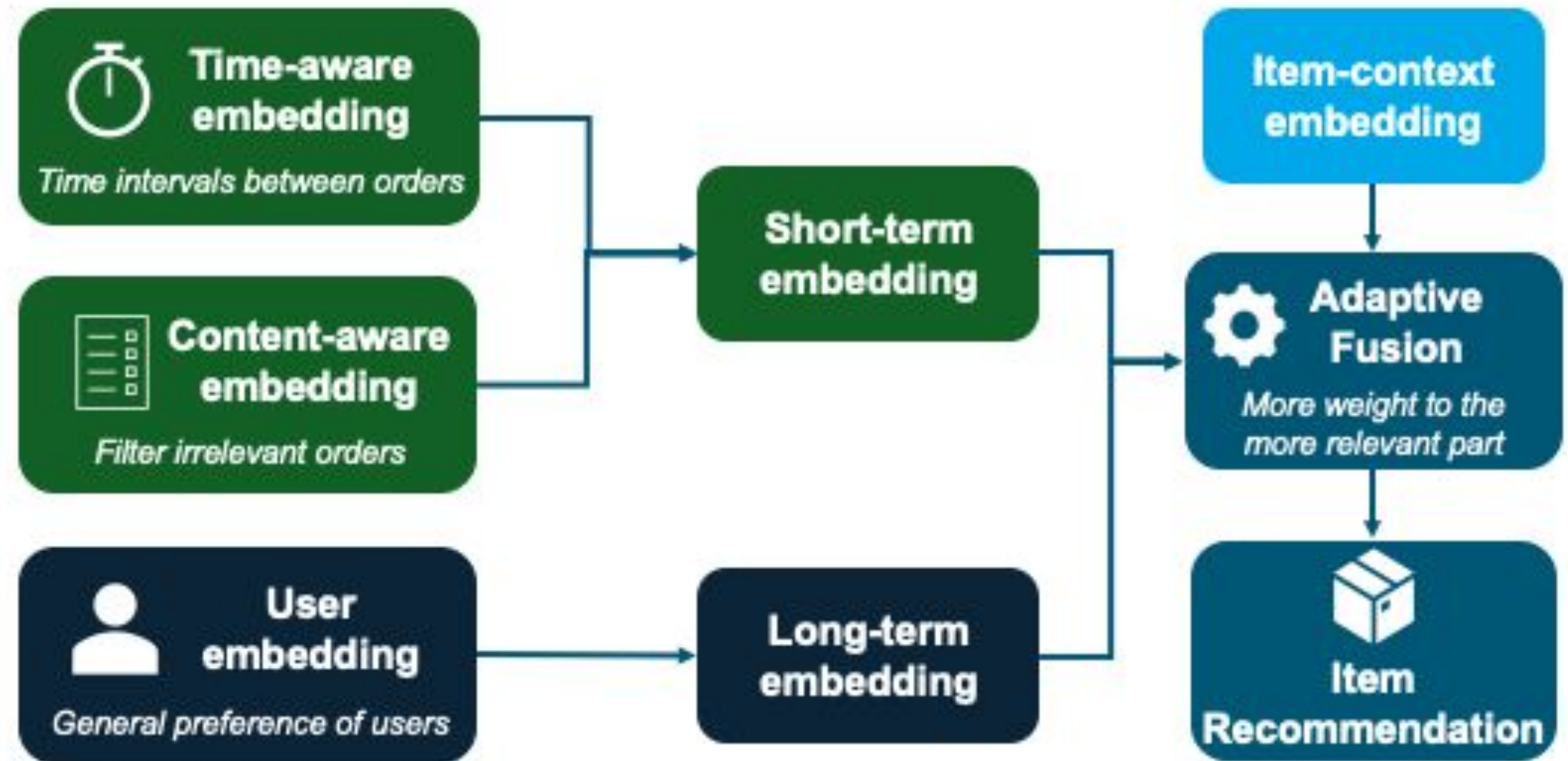
SLI-REC

Short term and Long term preference

Integrated RECommender system

(MICROSOFT)

1. Short-term activities
2. Long-term preferences
3. Adaptive combination



Who, when, or how much?

The generalization performance under different data sampling



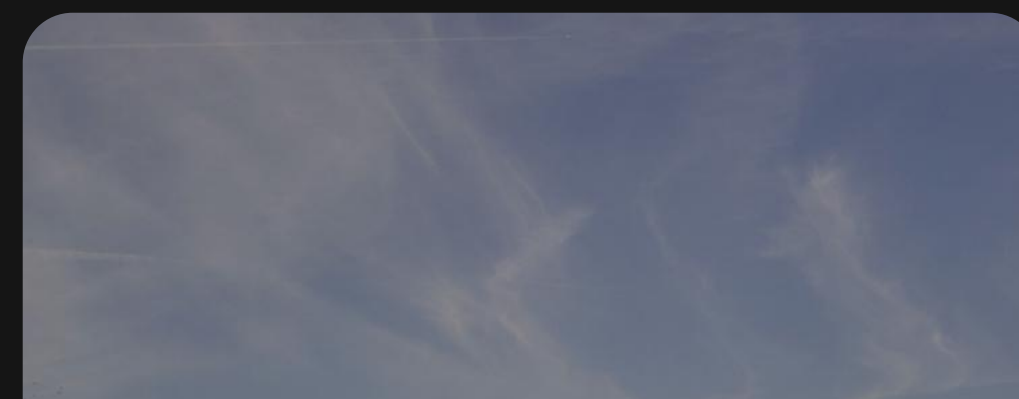
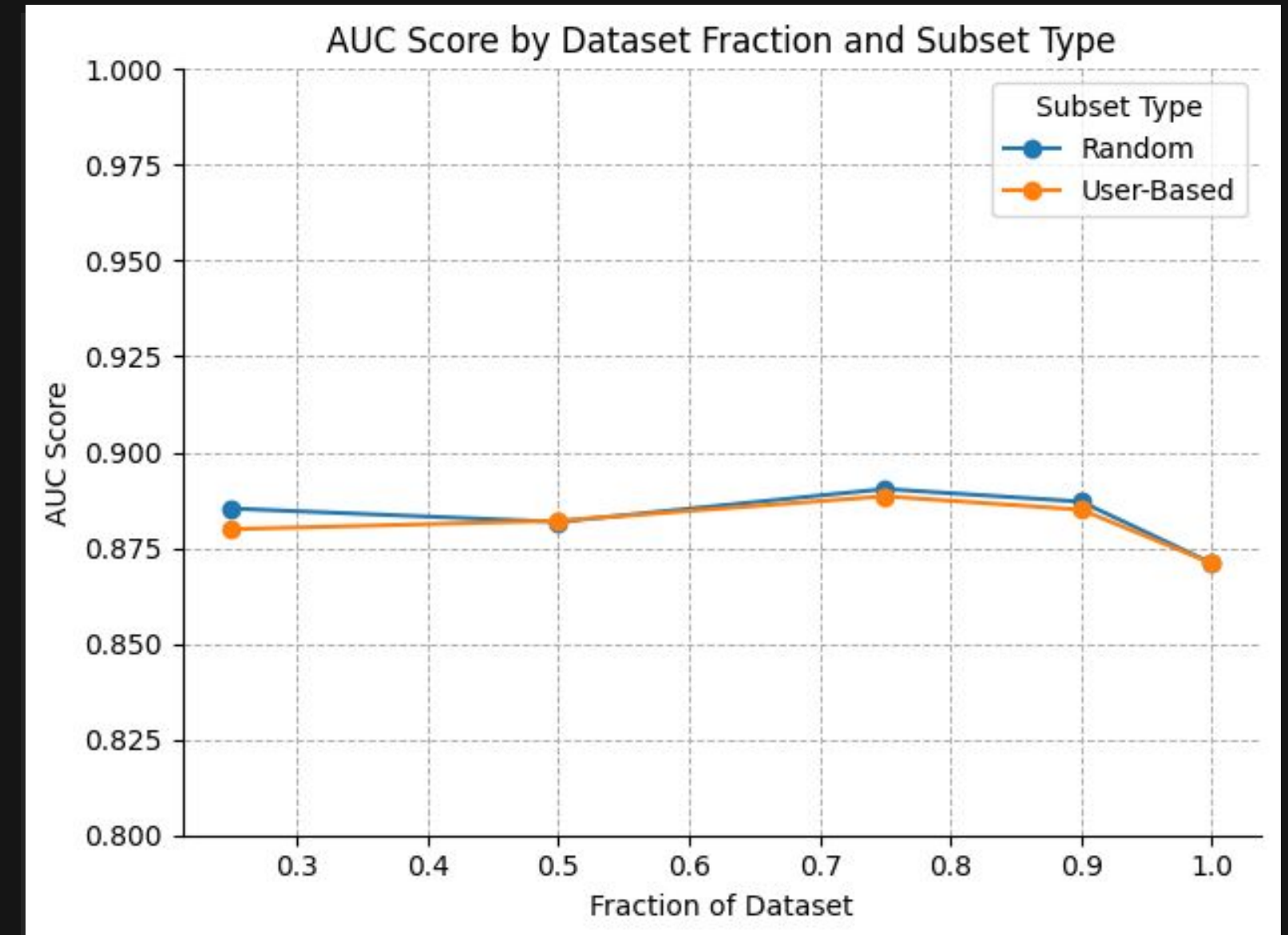
No. 1

User



No. 2

Random





#2 Question

You trained SLi-Rec on all the movies and TV shows you've binged. Now it's time to predict your next grocery run. How does it do?

a. As good as movies

b. Better than movies

c. Not great :(



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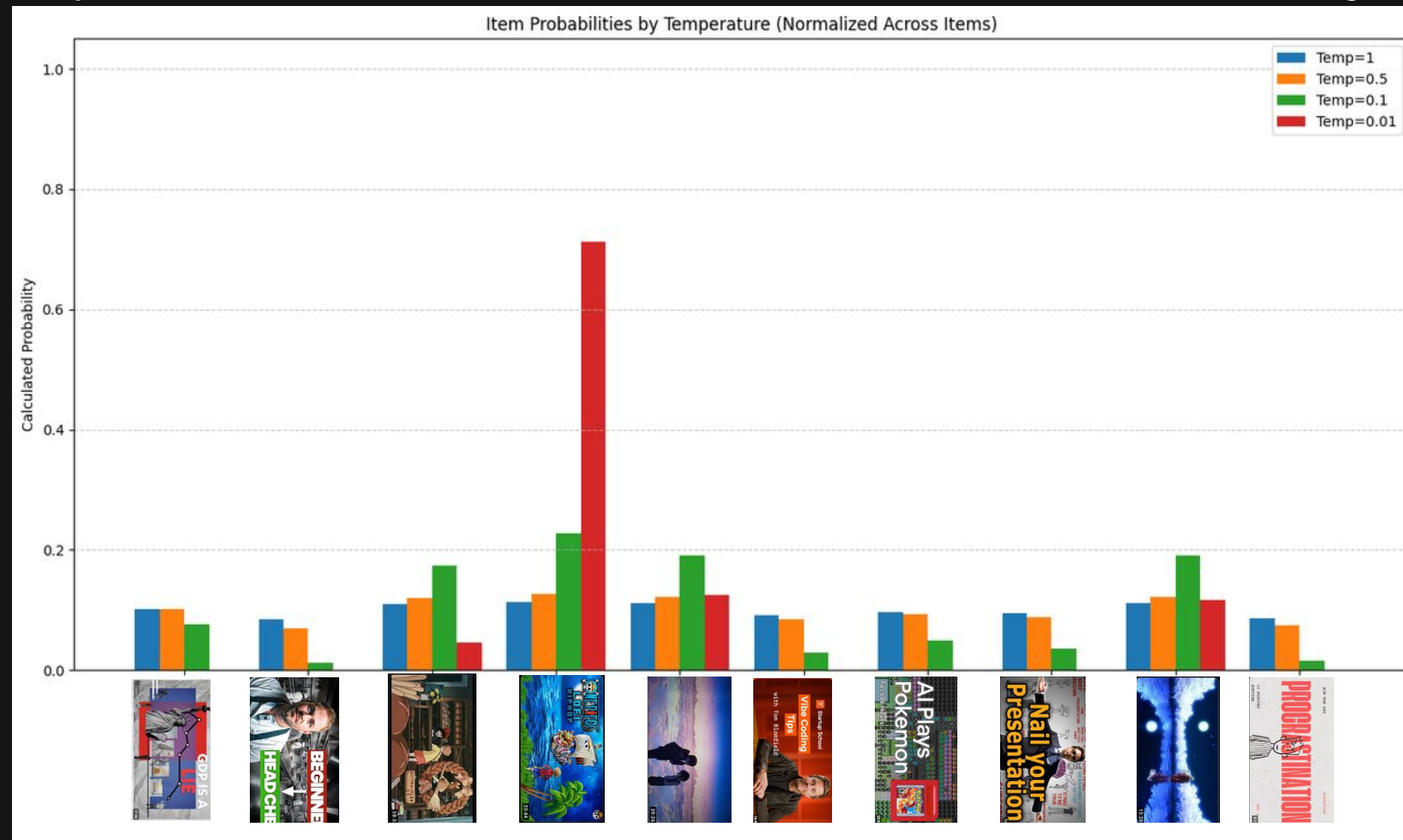
So which datasets do well?



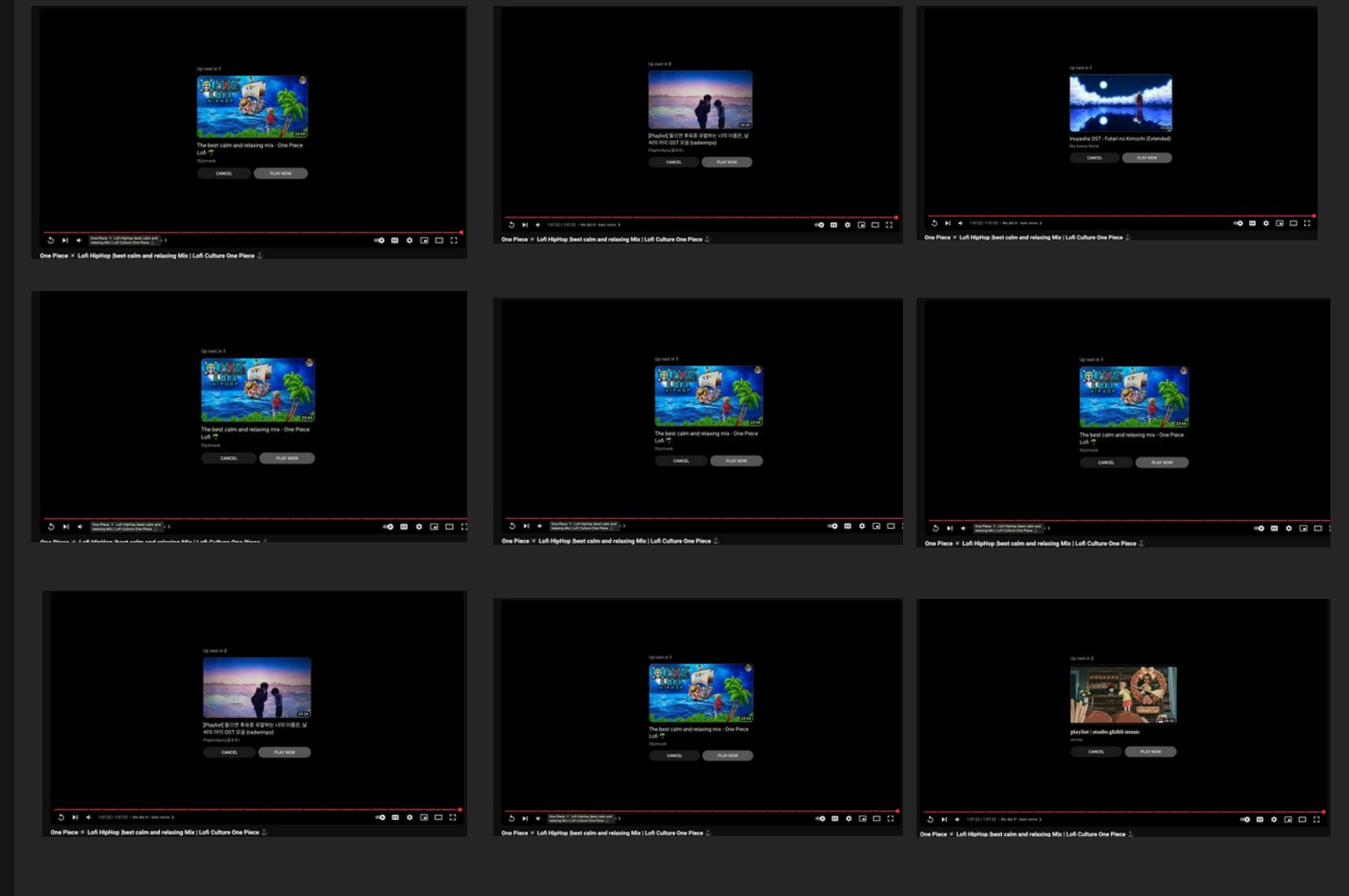
- SLi-Rec performs poorly out-of-the-box and requires retraining for effective use.
- SLi-Rec performs better on bigger and denser datasets

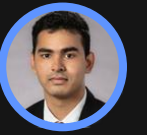
Is there something new?

Randomness on results reduce performance on 7%, but increases variety



A day of recommendations, Youtube:



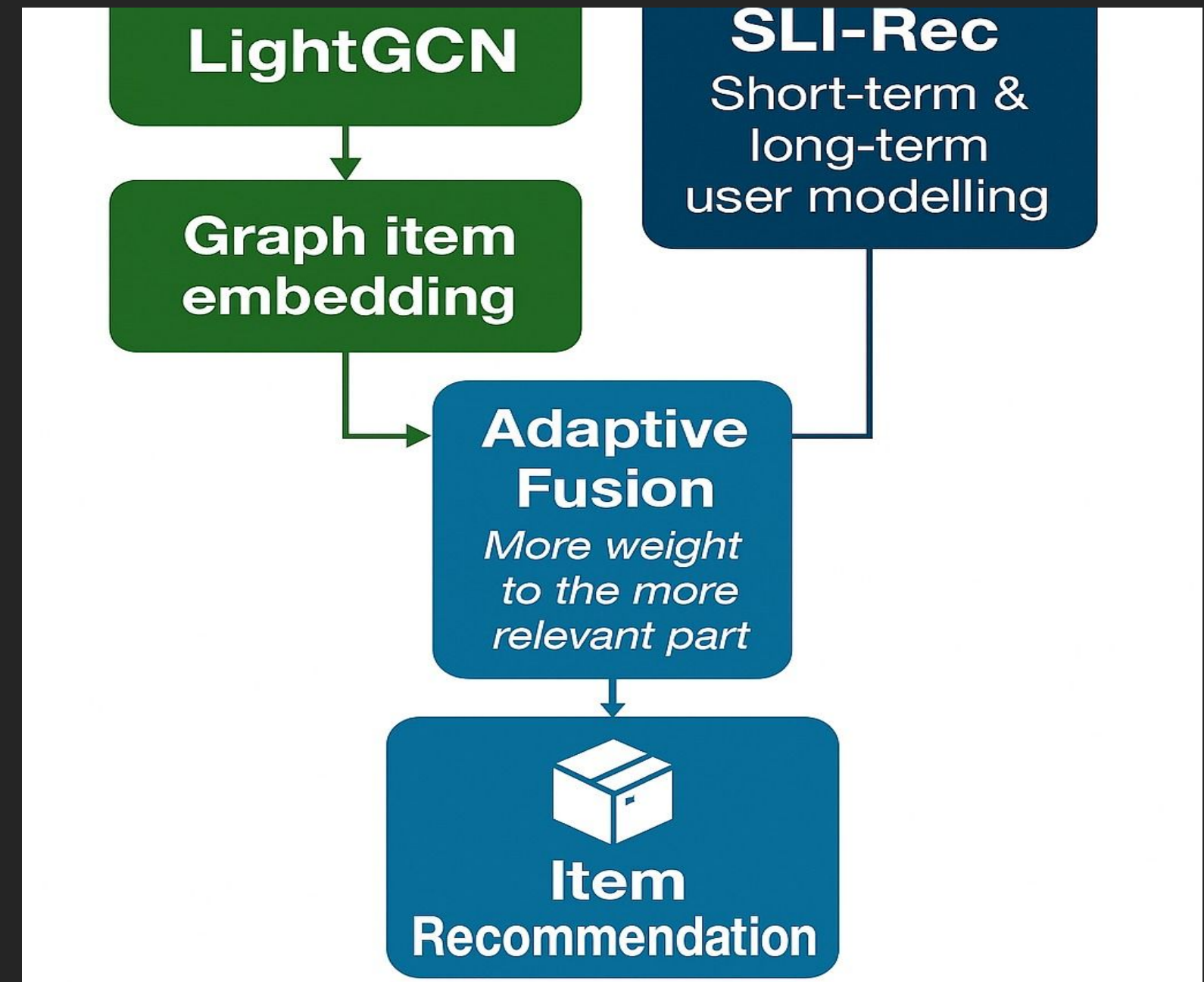


🔍 Something similar then?

We can enhance the model including a richer item context

- Captures deeper item relationships using a **graph**
- Works even with **limited or stale user history**
- Up to **37% improvement** in ranking accuracy
- Over **45% earlier hits** in relevance

Context Aware Sli-Rec



Recap!



No. 1

Recommendation systems are:

1. Time Sensitive
2. Common picks
3. Item context



No. 2

SLi-Rec Model is a great DL standard choice because it has all 3 components in a basic structure



No. 3

We found:

1. Has great generalization
2. Poor Out-of-box transfer
3. Adaptable to different needs



THANKS FOR LISTENING

